

Remote Data Visualization

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1. Course Content

1. Connect to the robot via a local area network web interface for remote data visualization and debugging.

[!TIP]

- When you need to observe and debug the robot's odometry, IMU, and other data, you can use the `foxglove` web interface to directly observe and plot the data, which is very important for debugging and visualizing the robot's data.

2. Preparations

2.1 Content Description

This lesson uses the Jetson Orin NX as an example. For Raspberry Pi boards, you need to open a terminal on the host machine, enter the command to get into the Docker container, and then input the commands mentioned in this lesson in the terminal within the Docker container. A tutorial on how to enter the Docker container from the host can be found in this product's guide [00.Configuration and Operation Guide]-[3.Enter the Docker (For RPI5)]. For ORIN and RDK boards, you can directly open a terminal and enter the commands mentioned in this lesson.

2.2 Start the Agent

Note: If it is already running, you do not need to start it again.

Enter the command in the vehicle's terminal:

```
start_agent
```

The terminal will print the following information, indicating a successful connection:

```
mircoROS_Agent
[1764926699.216025] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x2D836889, datareader_id: 0x000(6), subscri
ber_id: 0x000(4)
[1764926699.219768] info | ProxyClient.cpp | create_topic | t
topic created | client_key: 0x2D836889, topic_id: 0x006(2), participant_
id: 0x000(1)
[1764926699.222973] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x2D836889, subscriber_id: 0x001(4), partici
pant_id: 0x000(1)
[1764926699.227671] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x2D836889, datareader_id: 0x001(6), subscri
ber_id: 0x001(4)
[1764926699.232097] info | ProxyClient.cpp | create_topic | t
topic created | client_key: 0x2D836889, topic_id: 0x007(2), participant_
id: 0x000(1)
[1764926699.235075] info | ProxyClient.cpp | create_subscriber | s
subscriber created | client_key: 0x2D836889, subscriber_id: 0x002(4), partici
pant_id: 0x000(1)
[1764926699.239859] info | ProxyClient.cpp | create_datareader | d
datareader created | client_key: 0x2D836889, datareader_id: 0x002(6), subscri
ber_id: 0x002(4)
```

3. Run the Example

3.1 Start the Server Program

For Raspberry Pi 5 users, if the ROS container is not started, you need to start it first,

```
bringup_m3
```

Open a terminal inside the container,

```
exec_m3
```

- Start the server on the vehicle:

```
ros2 launch foxglove_bridge foxglove_bridge_launch.xml port:=8765
```

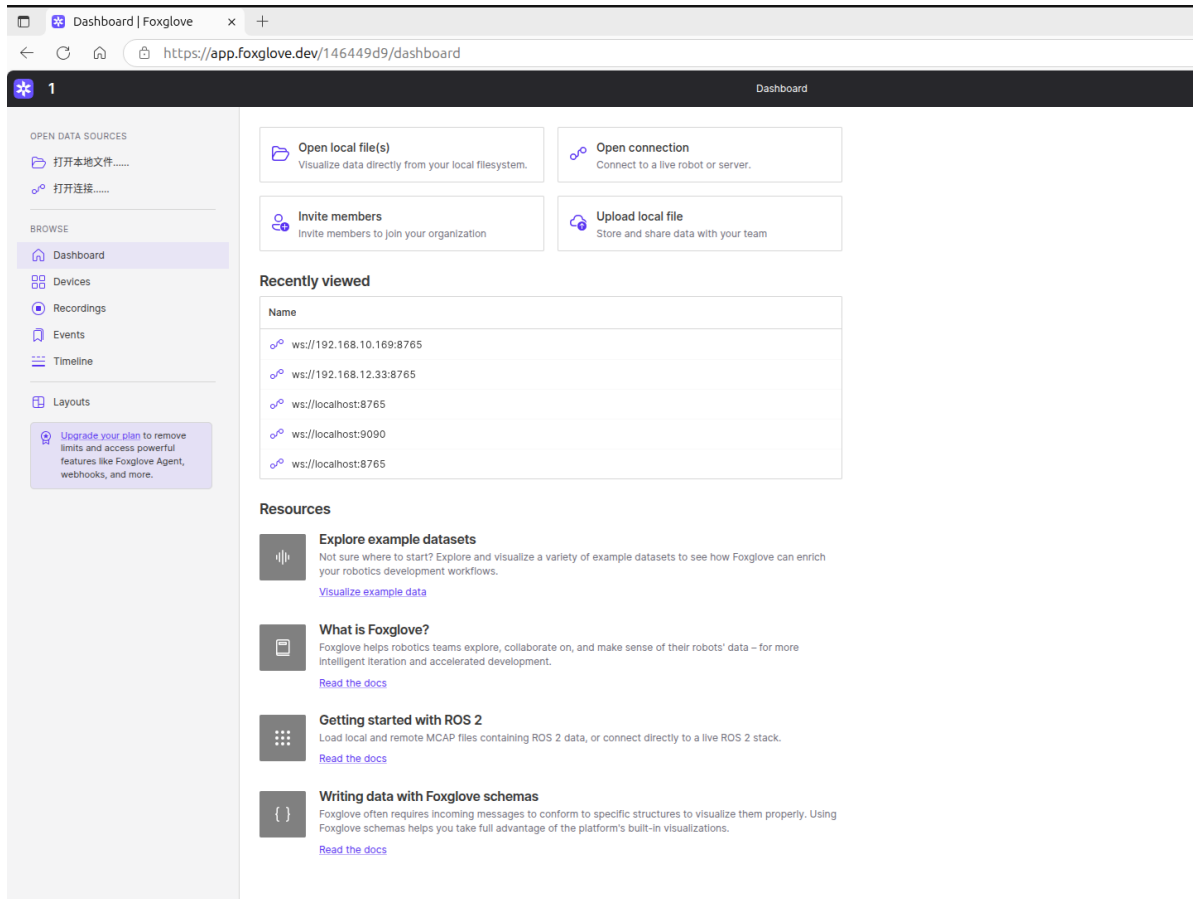
3.2 Remote Login

- Enter the following address in the browser's address bar (the vehicle and the PC must be on the same local network):

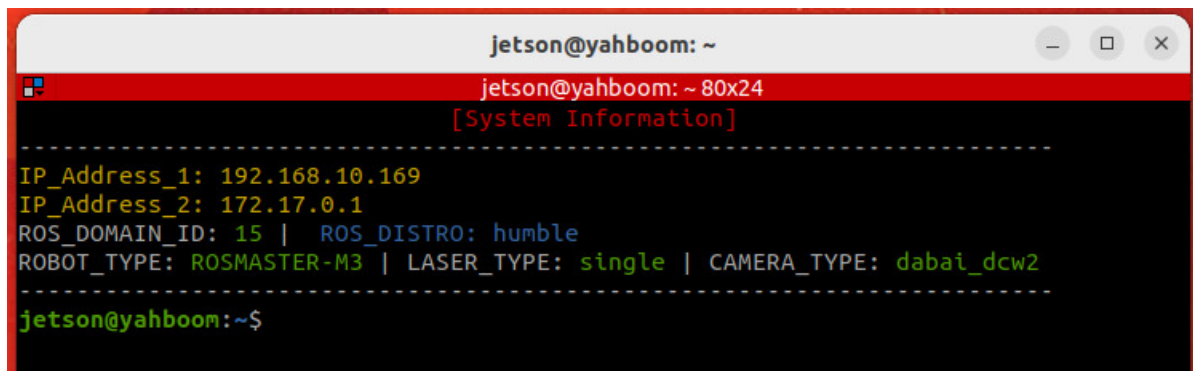
```
https://app.foxglove.dev/
```

[!TIP]

- If you are using it for the first time, you need to register a Foxglove account according to the page prompts.



- Open a terminal to check the current IP address of the vehicle, or you can observe it on the OLED screen in front of the vehicle.



- Click `open connection`.

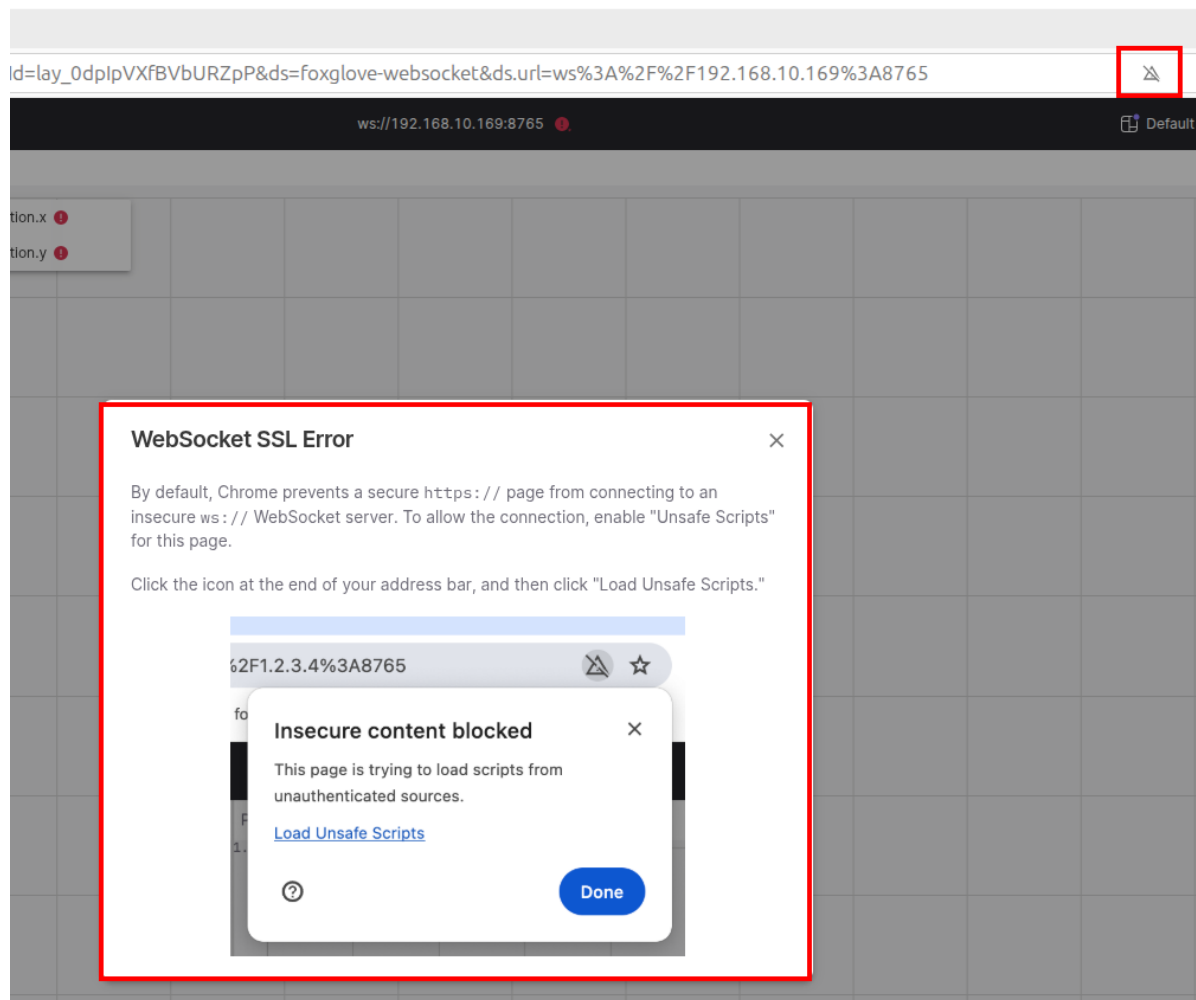
The screenshot shows the Foxglove web interface. On the left is a sidebar with navigation options: 'OPEN DATA SOURCES' (with buttons for '打开本地文件.....' and '打开连接.....'), 'BROWSE' (with links to 'Dashboard', 'Devices', 'Recordings', 'Events', 'Timeline', and 'Layouts'), and a 'Upgrade your plan' notification. The main area contains three action buttons: 'Open local file(s)', 'Open connection' (highlighted with a red border), and 'Upload local file'. Below these is a 'Recently viewed' table with five entries, all starting with 'ws://'. At the bottom is a 'Resources' section.

Name
ws://192.168.10.169:8765
ws://192.168.12.33:8765
ws://localhost:8765
ws://localhost:9090
ws://localhost:8765

- Here, we enter the robot's IP address + port 8765 in the URL.

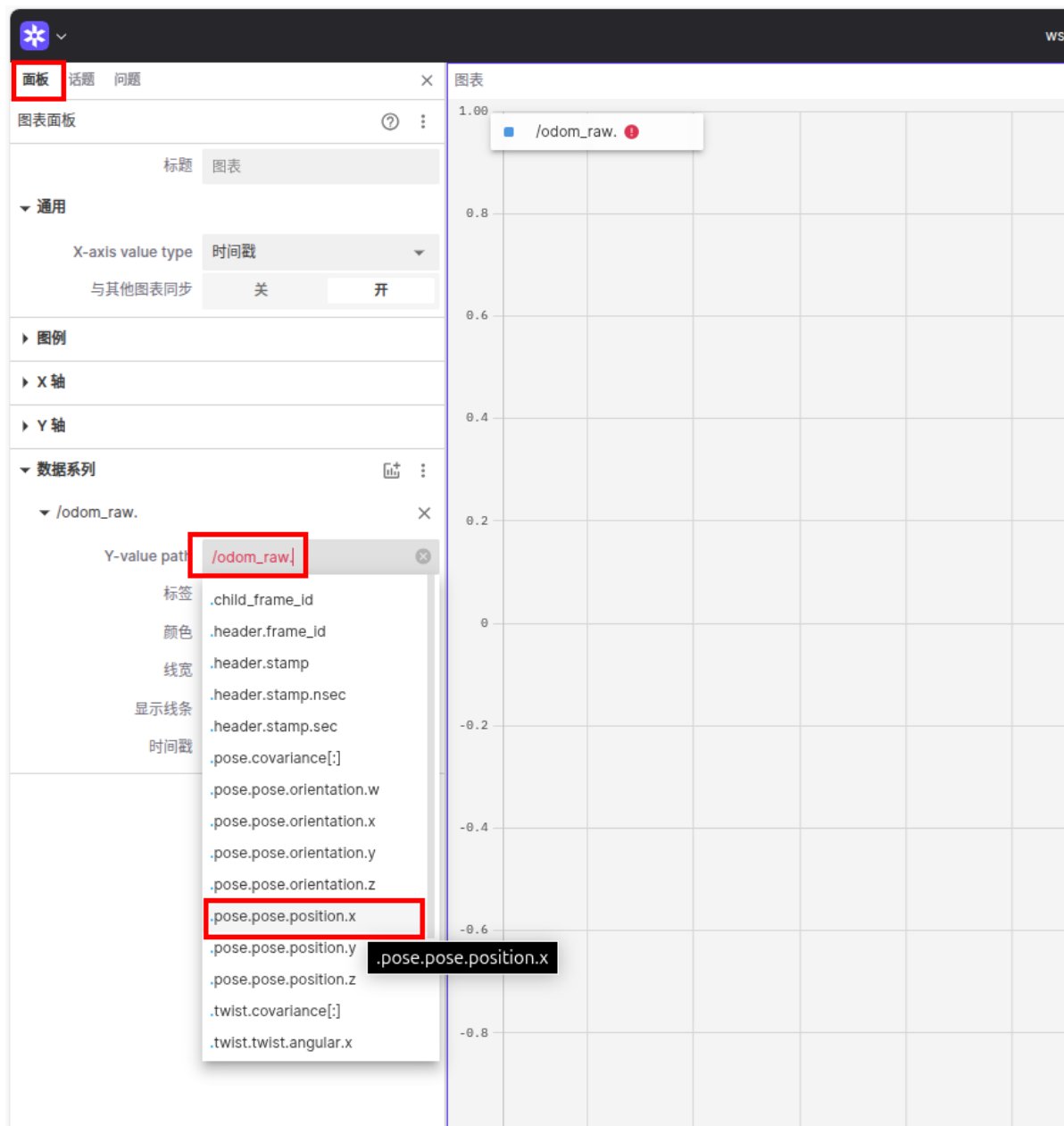
The screenshot shows a modal dialog titled 'Open a new connection'. On the left is a list of connection types: 'Foxglove WebSocket' (selected), 'Rosbridge', 'Remote files', 'ROS 1', 'ROS 2', and 'Velodyne Lidar'. The main area contains instructions: 'Connect to a ROS 1, ROS 2, or custom system using the Foxglove WebSocket protocol. For ROS systems, be sure to first install the foxglove_bridge ROS package.' Below this is a 'WebSocket URL' label and a text input field containing 'ws://192.168.10.169:8765', which is highlighted with a red border. At the bottom are links for 'View docs for ROS 1', 'View docs for ROS 2', and 'View docs for custom data', along with 'Cancel' and 'Open' buttons.

- If you see this prompt, it means that some browsers have a security block. Click the triangle icon to allow loading unsafe scripts.

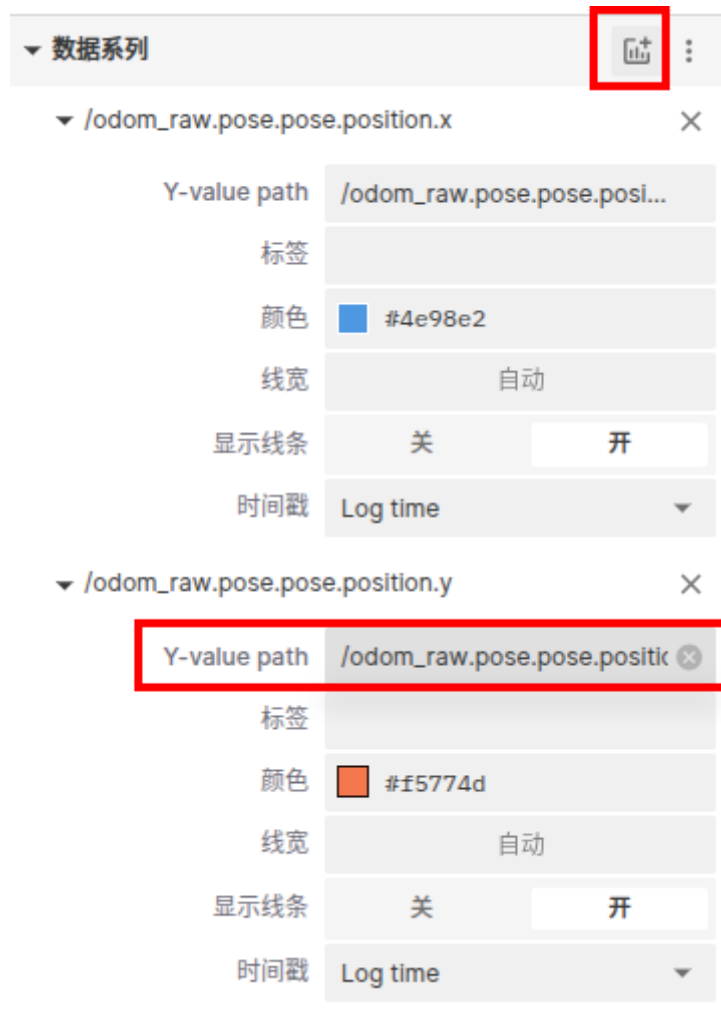


3.3 Visualize and Plot Odometry Data

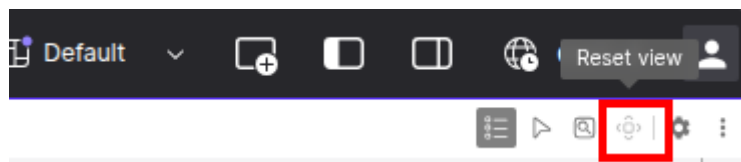
- Here, we will take plotting and observing the planar position changes of the vehicle's wheel odometry as an example.
- Click on the left panel -> `Data Series` -> `/odom_raw` -> `pose.pose.position.x`.



- Then click the **+** sign to add a new data item.
- Following the previous step, add `/odom_raw` -> `pose.pose.position.y`.



- Click **Reset view** in the upper left corner to reset the view.



- At this point, you can see that there is already data.



- Start the keyboard control node to move the vehicle and observe the data changes.

keyboard_control

```
jetson@yahboom:~$ keyboard_control

This node takes keypresses from the keyboard and publishes them
as Twist/TwistStamped messages. It works best with a US keyboard layout.
-----
Moving around:
  u   i   o
  j   k   l
  m   ,   .

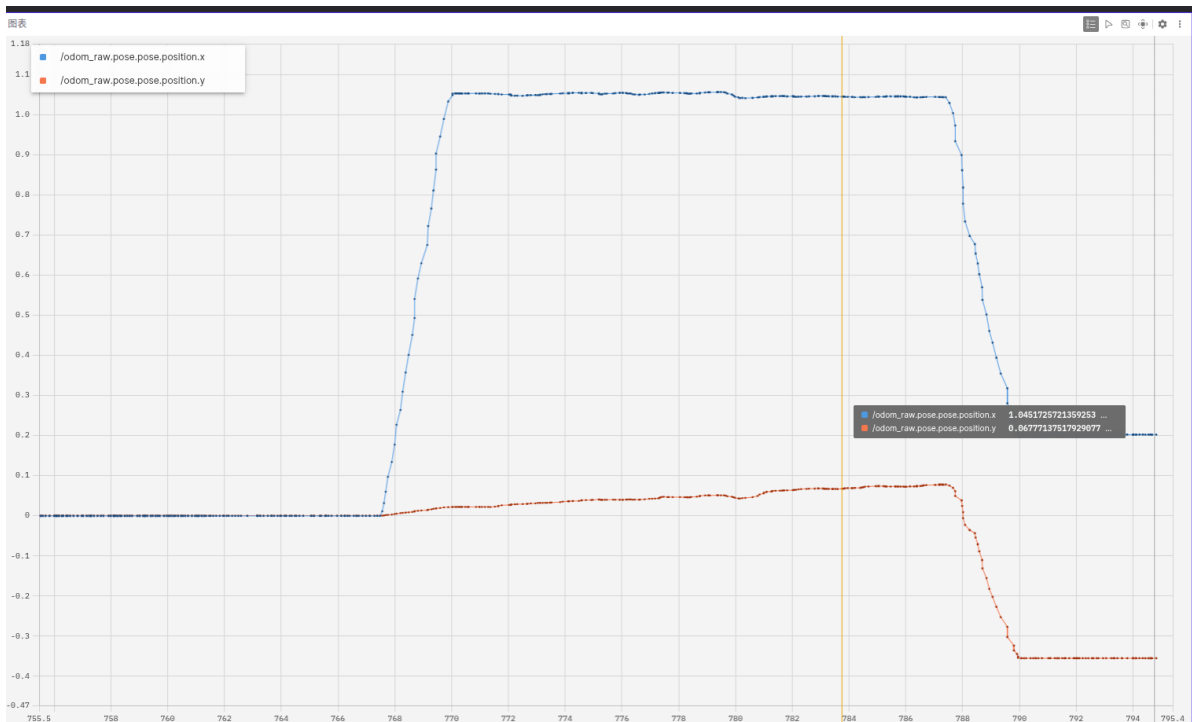
For Holonomic mode (strafing), hold down the shift key:
-----
  U   I   O
  J   K   L
  M   <   >

t : up (+z)
b : down (-z)

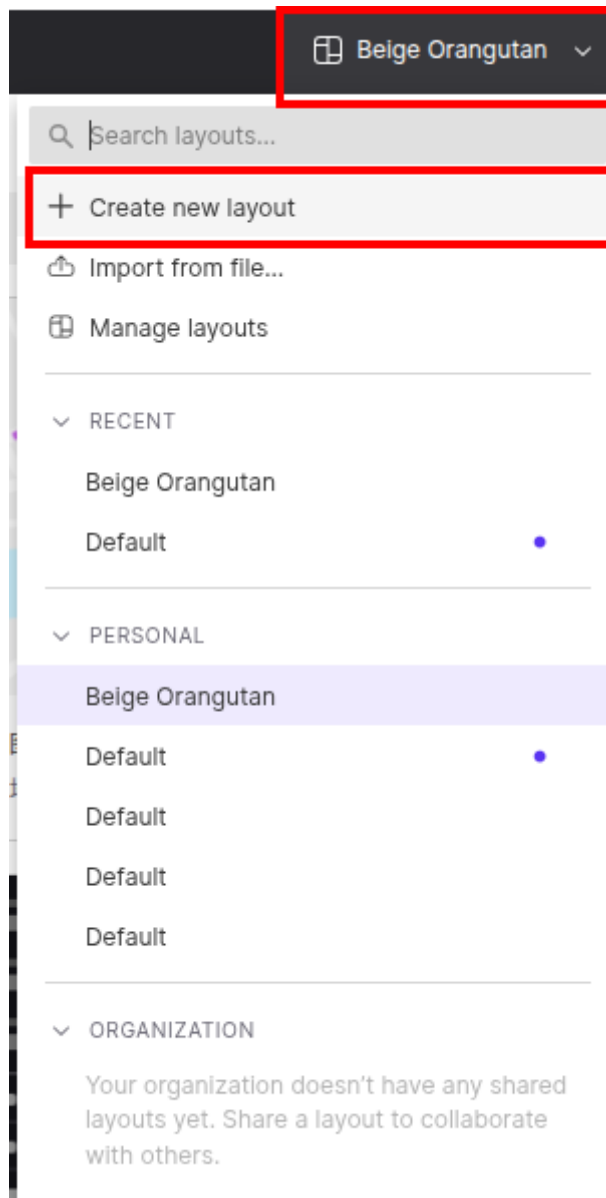
anything else : stop

q/z : increase/decrease max speeds by 10%
w/x : increase/decrease only linear speed by 10%
```

- Now we can see the detailed changes in the robot's position on the X and Y axes in the plane. You can zoom in and out by hovering the mouse over the X/Y axes.



- You can also create a new display layout by clicking **Create new layout**, with various types of data to choose from.



- Here, you can choose and add visualization panels according to your needs. We will not demonstrate them one by one here.

